

Thermal Performance tests on Three Radiators

Multi Column Types

Carried out for
Tianjin Jiuding Yangguang HVAC Co., Ltd.

Report 105549/1

Compiled by Tomas Webster

21 December 2023



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Multi Column Types

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Contract: Report 105549/1

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QUALITY ASSURANCE

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SUMMARY

Thermal performance tests were carried out on three radiators in accordance with the procedures contained in BS EN 442-2:2014, using low pressure hot water in the BSRIA radiator test room, described within Section 3 of this report. The work was requested by Tianjin Jiuding Yangguang HVAC Co., Ltd. and was carried out during the period 15 – 27 November 2023. Pressure testing was conducted on 20 December 2023.

A summary of the thermal performance is shown below.

Model	Test reference	Width	Height	Columns	Output W at 30ΔT	Output W at 50ΔT	Factor 'Km'	Exponent 'n'
6012	105549A1TW	410	1805	6 x 2	627	1174	9.8546	1.2291
4020	105549A2JG	284	1599	6 x 2	390	743	5.3253	1.2624
22K	105549A3TW	1000	602	30	808	1570	9.7158	1.2999

Full test results are contained within section 5 of this report.

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1 INTRODUCTION

Thermal performance tests were carried out on three radiators in accordance with the procedures contained in BS EN 442-2:2014, using low pressure hot water in the BSRIA radiator test room, described within Section 3 of this report. The work was requested by Tianjin Jiuding Yangguang HVAC Co., Ltd. and was carried out during the period 15 – 27 November 2023. Pressure testing was conducted on 20 December 2023.

All samples were received on 29 September 2023. All samples were received in good condition.

This report relates to the samples tested and no others.

Comments and opinions are outside the scope of UKAS accreditation.

2 ITEMS RECEIVED FOR TEST

The test samples consisted of two Multi Column Types and one panel radiator from Tianjin Jiuding Yangguang HVAC Co., Ltd.

The samples were designated by the manufacturer as follows:

6012
4020
22K

The samples consisted of two multi-column radiators and one panel radiator. Samples A1 and A3 were plumbed top and bottom same end (T.B.S.E.) with the internal blocker removed from sample A1. Sample A2 was plumbed bottom opposite ends (B.O.E.).

The radiators were installed as standard for BS EN 442 but using the supplied mountings for spacing from the test chamber wall.

Further details of the test samples can be found in Appendix A, with manufacturer's drawings included in Appendix B to enable the sample types, sizes, test pressures and finish to be verified.

3 TEST FACILITY

The test facility consists of a test room 4.0m (l) x 4.0m (w) x 3.0m (h), which is constructed to the requirements contained in BS EN 442-2:2014, i.e., five water cooled surfaces and one insulated surface against which the test radiator is installed.

When steady state conditions are achieved the appliance output is determined from measurements of the water flow rate and inlet/outlet water temperature difference.

A standard test consists of three test points. For all appliances a first test is carried out with water supply temperature that produces a mean water temperature of 70°C with the water flow rate such that for radiators the inlet/outlet water temperature difference is 10°C. Two further tests are carried out at the water flow rate established in the first test but with different supply water temperatures.

For products marked¹ the temperature drop was set at 5°C to enable the flow rate to remain within the standard measurement uncertainty for the test chamber.

For all tests the enclosure air temperature is controlled to maintain 20°C at the inner room reference point, which is 0.75m from the floor in the centre of the room.

4 INSTRUMENTATION

TEST RECORD SHEET TP21/7 : TEST EQUIPMENT / INSTRUMENTS

	Instrument No.	Calibration expiry date
Weigh scales	2026, 2027	16/01/2024
Resistance thermometer (air) reference & radiant shield	329	07/03/2024
Resistance thermometers (water) reference	1512-1515	07/03/2024
Digital voltmeter 7½ digit (resistance thermometer measurement)	266	07/03/2024
Barometer	881	02/12/2023
Electronic timer within PC	526	24/01/2024
Vernier caliper (0 – 1.5m)	359	22/03/2025
Digital caliper (0 – 150mm)	2057	05/07/2024
Weigh scales (water content)	188	17/01/2024

Comments: None

Test Engineer

Tomas Webster

5 THERMAL RESULTS

Model	Test reference	Width	Height	Columns	Output W at 30ΔT	Output W at 50ΔT	Factor 'Km'	Exponent 'n'
6012	105549A1TW	410	1850	6 x 2	627	1174	9.8546	1.2291
4020	105549A2JG	284	1549	6 x 2	390	743	5.3253	1.2624
22K	105549A3TW	1000	602	30	808	1570	9.7158	1.2999

Results for the products are for those heights tested, enabling catalogues to be generated based on overall linear length for each height tested.

Catalogue outputs can be derived for other lengths of those heights tested by using a linear length constant of 1 (divide test output by test length for relevant height and multiply by desired length).

Test data is valid for the type, material and surface finish tested. The data may also be valid for other surface finishes with the same emissivity.

Individual test results for samples in this contract are shown below.

Date of Test: 27/11/2023
 Manufacturer: Tianjin
 Model Reference: 6012
 Test Reference Number: 105549A1TW
 Type of Heater: Multi Column
 Surface Finish: Anthracite Coat
 Pipework Connections: T.B.S.E.

HEATER DIMENSIONS

Overall Height: (mm) 1805
 Overall Length: (mm) 410
 Overall Depth: (mm) 60
 Convector Height: (mm) N/A
 Convector Depth: (mm) N/A
 Height above floor: (mm) 110
 Distance from wall: (mm) 29

Radiated heat factor Sk 0.30
 Barometer exponent np 0.50

MEAN TEST VALUES	TEST 1	TEST 2	TEST 3
Room air temperature 0.75m: (°C)	19.5	19.5	19.5
Flow rate (g/s)	30.8	30.6	30.5
Flow enthalpy (J/g)	314.5	215.7	355.5
Return enthalpy (J/g)	274.8	195.8	308.8
Flow temperature (°C)	75.2	51.5	84.9
Return temperature (°C)	65.7	46.8	73.8
Output (W)	1224.6	608.7	1426.9
Mean water temperature (°C)	70.4	49.2	79.4
Temperature difference (°C)	50.9	29.6	59.8
Barometric pressure (mbar)	988.6	990.4	1002.5
Corrected output (W)	1235.2	613.5	1432.1
Estimated output (W)	1200.6	617.6	1463.8

PERFORMANCE EQUATION

$$\text{Output (W)} = K_M (\text{mean water temperature} - \text{room air temperature})^n$$

From test results $K_M = 9.5846$
 $n = 1.2291$

VARIATION OF OUTPUT WITH TEMPERATURE DIFFERENCE

TEMPERATURE DIFFERENCE °C	HEAT OUTPUT W
20	381
30	627
40	893
50	1174
60	1469
70	1776

Date of Test: 15/11/2023
 Manufacturer: Tianjin
 Model Reference: 4020
 Test Reference Number: 105549A2JG
 Type of Heater: Multi Column
 Surface Finish: Anthracite Coat
 Pipework Connections: B.O.E.

HEATER DIMENSIONS

Overall Height: (mm) 1599
 Overall Length: (mm) 298
 Overall Depth: (mm) 70
 Convector Height: (mm) N/A
 Convector Depth: (mm) N/A
 Height above floor: (mm) 110
 Distance from wall: (mm) 32

Radiated heat factor Sk 0.30
 Barometer exponent np 0.50

MEAN TEST VALUES	TEST 1	TEST 2	TEST 3
Room air temperature 0.75m: (°C)	20.0	20.2	20.0
Flow rate (g/s)	17.9	17.9	17.8
Flow enthalpy (J/g)	311.8	222.4	360.0
Return enthalpy (J/g)	270.2	200.4	308.4
Flow temperature (°C)	74.5	53.1	86.0
Return temperature (°C)	64.6	47.9	73.7
Output (W)	744.4	393.5	920.5
Mean water temperature (°C)	69.5	50.5	79.8
Temperature difference (°C)	49.5	30.3	59.8
Barometric pressure (mbar)	1009.1	1009.5	1010.0
Corrected output (W)	745.4	394.0	921.4
Estimated output (W)	734.6	395.6	931.3

PERFORMANCE EQUATION

$$\text{Output (W)} = K_M (\text{mean water temperature} - \text{room air temperature})^n$$

From test results $K_M = 5.3253$
 $n = 1.2624$

VARIATION OF OUTPUT WITH TEMPERATURE DIFFERENCE

TEMPERATURE DIFFERENCE °C	HEAT OUTPUT W
20	234
30	390
40	561
50	743
60	935
70	1136

Date of Test: 20/11/2023
 Manufacturer: Tianjin
 Model Reference: 22K
 Test Reference Number: 105549A3TW
 Type of Heater: Double Panel Double Convector
 Surface Finish: White Powder Coat
 Pipework Connections: B.O.E.

HEATER DIMENSIONS

Overall Height: (mm) 602
 Overall Length: (mm) 1000
 Overall Depth: (mm) 100
 Convector Height: (mm) 530
 Convector Depth: (mm) 32
 Height above floor: (mm) 110
 Distance from wall: (mm) 40

Radiated heat factor Sk 0.20
 Barometer exponent np 0.70

MEAN TEST VALUES	TEST 1	TEST 2	TEST 3
Room air temperature 0.75m: (°C)	20.0	19.8	20.1
Flow rate (g/s)	37.9	38.1	37.8
Flow enthalpy (J/g)	315.2	216.0	359.5
Return enthalpy (J/g)	273.6	195.5	307.7
Flow temperature (°C)	75.3	51.6	85.9
Return temperature (°C)	65.4	46.7	73.5
Output (W)	1573.8	782.2	1959.3
Mean water temperature (°C)	70.3	49.2	79.7
Temperature difference (°C)	50.4	29.4	59.5
Barometric pressure (mbar)	1000.6	1001.1	1002.1
Corrected output (W)	1584.7	787.4	1971.2
Estimated output (W)	1585.6	787.3	1970.3

PERFORMANCE EQUATION

$$\text{Output (W)} = K_M (\text{mean water temperature} - \text{room air temperature})^n$$

From test results $K_M = 9.7158$
 $n = 1.2999$

VARIATION OF OUTPUT WITH TEMPERATURE DIFFERENCE

TEMPERATURE DIFFERENCE °C	HEAT OUTPUT W
20	477
30	808
40	1175
50	1570
60	1990
70	2432

6 PRESSURE TESTS

TEST RECORD SHEET TP21/4 : BURST TEST

BS EN 442-1 : 2014 clause 5.6 requirements

Sample heating appliances shall be subjected to a burst test at a pressure 1.69 times the maximum working pressure (See commentary below).

- The sample under test may deform but shall not rupture.
- The sample radiators shall not be less than 500mm long.

Commentary

The leak testing pressure is specified by the manufacturer (usually on the product drawing) and is at least 1.3 times the quoted maximum operating pressure. This pressure shall not be less than 520 kPa (75.4 psi). BS EN 442-1 : 2014 clause 5.4 refers. Furthermore, the burst test pressure shall be maintained for a minimum period of 2 minutes.

Test instrumentation

	BSRIA instrument number	Calibration expiry date
Standard pressure gauge	1535	16/05/2024
Stopwatch	183	20/01/2024

Test data

Test item reference number	Date of Test	Burst test pressure [1.3 x leak test pressure] (Bar)	Did test item hold burst pressure for a minimum of 2min without rupture Pass/Fail
105549A1TW	20/12/2023	6.76	Pass
105549A2JG	20/12/2023	16.9	Pass
105549A3TW	20/12/2023	16.9	Pass

TEST ENGINEER

Tomas Webster

APPENDIX A: PRODUCT INFORMATIONTEST RECORD SHEET TP21/1: TEST ITEMS

Contract number		105549A	
Date of receipt	Test Engineer initials	Full description of test item	Test item reference number
29/09/2023	TW	Steel Anthracite multi column radiator with 6 rows, 1800x405	105549A1TW
29/09/2023	JG	Steel Anthracite multi column radiator with 6 rows, 1600x297	105549A2JG
29/09/2023	TW	Steel White panel radiator with 2 panels, 2 convectors, 602x1000	105549A3TW

Comments:- Test item reference numbers list all items received in the numerical order they were tested.

Test Engineer (signature)

Tomas Webster

TEST RECORD SHEET TP21/2 : PRODUCT INFORMATION

BSRIA test reference number	105549A1TW
Client	Tianjin Jiuding Yangguang HVAC Co., Ltd.
Manufacturer	Tianjin Jiuding Yangguang HVAC Co., Ltd.
Product reference number	6012
Product style	Multi Column
Material of construction	Steel Anthracite Coat
Date of receipt	29/09/2023
Product or packaging markings	N/A
Test start date	09/10/2023
Weight (dry) (kg)	31.5
Water content (kg)	8.96

DIMENSIONAL MEASUREMENTS

Measurement Parameter	Measured value (mm)	Manufacturer's stated value (mm)	EN 442-2 dimensional tolerance (mm or %)	Pass / Fail
Overall height	1805	1800	+6/-2	Pass
Overall length	410	405	+/-1.5%	Pass
Overall depth	60	61	+/-1.5	Pass
Convactor height	N/A			
Convactor depth	N/A			

Number of columns per panel	6 x 2
Distance installed from the wall (mm)	29
Distance between centres (mm)	1765
Panel thickness (mm)	13.34
Convactor overall length (mm)	N/A
Convactor thickness (mm)	N/A
Spot weld horizontal pitch (mm)	7.95
Additional information	B.O.E

TEST ENGINEER (Signature)

Tomas Webster

TEST RECORD SHEET TP21/2 : PRODUCT INFORMATION

BSRIA test reference number	105549A2JG
Client	Tianjin Jiuding Yangguang HVAC Co., Ltd.
Manufacturer	Tianjin Jiuding Yangguang HVAC Co., Ltd.
Product reference number	4020
Product style	Multi Column
Material of construction	Anthracite Coat
Date of receipt	29/09/2023
Product or packaging markings	N/A
Test start date	15/11/2023
Weight (dry) (kg)	22.2
Water content (kg)	9.17

DIMENSIONAL MEASUREMENTS

Measurement Parameter	Measured value (mm)	Manufacturer's stated value (mm)	EN 442-2 dimensional tolerance (mm or %)	Pass / Fail
Overall height	1599	1600	+6/-2	Pass
Overall length	298	297	+/-1.5%	Pass
Overall depth	70	70	+/-1.5	Pass
Convactor height	N/A			
Convactor depth	N/A			

Number of columns per panel	6 x 2
Distance installed from the wall (mm)	56
Distance between centres (mm)	32
Panel thickness (mm)	21.05
Convactor overall length (mm)	N/A
Convactor thickness (mm)	N/A
Spot weld horizontal pitch (mm)	7.85
Additional information	B.O.E.

TEST ENGINEER (Signature)

Joseph George

TEST RECORD SHEET TP21/2 : PRODUCT INFORMATION

BSRIA test reference number	105549A3TW
Client	Tianjin Jiuding Yangguang HVAC Co., Ltd.
Manufacturer	Tianjin Jiuding Yangguang HVAC Co., Ltd.
Product reference number	22K
Product style	Multi Column
Material of construction	White Powder Coat
Date of receipt	29/09/2023
Product or packaging markings	N/A
Test start date	20/11/2023
Weight (dry) (kg)	30.8
Water content (kg)	5.85

DIMENSIONAL MEASUREMENTS

Measurement Parameter	Measured value (mm)	Manufacturer's stated value (mm)	EN 442-2 dimensional tolerance (mm or %)	Pass / Fail
Overall height	602	600	+4/-2	Pass
Overall length	1000	1000	+/-5	Pass
Overall depth	100	100	+4/-3	Pass
Convactor height	530			
Convactor depth	32			

Number of columns per panel	30
Distance installed from the wall (mm)	40
Distance between centres (mm)	547
Panel thickness (mm)	13.18
Convactor overall length (mm)	968
Convactor thickness (mm)	0.32
Spot weld horizontal pitch (mm)	32.3
Additional information	T.B.S.E.

TEST ENGINEER (Signature)

Tomas Webster

APPENDIX B: MANUFACTURERS PRODUCT INFORMATION





